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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 09/01/2006 |
| **PI Name:** | Fernandez-Pello, Carlos Ph.D. | | |
| **Project Title:** | Two-dimensional Smoldering and Its Transition to Flaming in Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 99-HEDS-04 |
| **Start Date:** | 01/29/2001 | **End Date:** | 05/31/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Urban, David L. ( NASA Glenn Research Ctr ) | | |
| **Grant/Contract No.:** | NCC3-874 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

The purpose of the program is to study smoldering and its transition to flaming in microgravity conditions. The experiment consists of a piece of the test fuel inserted in the wall of a flow duct. The back and sides of the sample are sealed and thermally insulated. The upstream end of the sample is heated by an igniter to initiate smolder propagation. The experimental measurements include the forward smolder propagation velocity and the smolder reaction temperature. The experimental parameters include the duct flow velocity, oxygen concentration, and external heat flux. The fuels to be studied include polyurethane foam, Pyrell stowage foam and other spacecraft materials.

The objectives of this research program are to conduct microgravity experiments on the transition to flaming, to incorporate the results of these tests and complementary normal gravity tests into a detailed physical model of the smoldering process and the transition to flaming, and to build a simplified model and test apparatus to evaluate the smoldering characteristics of a variety of materials. The specific objectives are:

**Task Description:**

1. To continue normal gravity testing on practical materials used in spacecraft (i.e., cellulose, Pyrell) to determine the effect of oxidizer flow velocity, oxygen concentration, and external heat flux on the transition from smoldering to flaming, using current and new diagnostic methods.

2. To apply new diagnostic techniques (i.e., ultrasound imaging, particle imaging velocimetry, mass loss, and gas sampling) to study smoldering of polyurethane foam, and then apply the new techniques to other materials.

3. To continue the technical support to the NASA STaF team to design and conduct microgravity experiments in the ISS on the transition from smoldering to flaming of polyurethane foam and other materials mentioned above.

4. To develop a numerical model of two-dimensional smoldering combustion and transition to flaming, solving for the governing equations with boundary conditions similar to those in the experimental apparatus. To support the modeling, thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC) experiments will be used to extract the kinetics and heats of reaction of each material. The model will be validated with the experiments and then used to predict smolder behavior of materials on earth and in space, avoiding the high cost of additional experiments.

5. To provide a new data bank for smoldering and transition to flaming. The database will be used to validate and verify present and future theoretical models.

6. To develop a standard smolder test that can provide material properties and parameters (i.e., critical heat flux, smolder temperature profiles, mass loss, species yields, and thermal properties) for ranking and evaluating the smoldering and fire risk for wide variety of materials. These parameters will be used in a simplified smolder model for further characterization of its smolder behavior at different conditions and configurations.

Smoldering is of interest not only as a serious fire risk, but also as a fundamental combustion problem. It encompasses a number of fundamental processes, including: heat and mass transfer in a porous media, endothermic and oxidative pyrolysis, ignition, heterogeneous exothermic reactions at the solid/gas pore interface, and the induction of gas phase reactions (flaming) from the existing surface reactions. These mechanisms are complex and the removal of gravity provides substantial simplifications to their study. Furthermore, there is currently only limited information of the behavior of smoldering and the transition to flaming in a space facility environment, and there is a need for further information for fire prevention and control in earth and space facilities. Space based experiments are necessary because smoldering is a very slow process and its study in microgravity requires periods of time that can only be achieved in space facilities.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Ultrasound Probing System
An ultrasound probing technique has been implemented into the test apparatus to measure the permeability of the sample at the same heights as the top five thermocouples. The ultrasound technique uses five speaker-microphone pairs to track the progress of the smolder reaction and to observe the evolution of the char, made possible because the permeability of the char is significantly larger than that of the virgin foam. The technique is based on the fact that the attenuation of an ultrasound signal through a porous medium is related to the permeability of the medium. In the present work, the output signal provides a relative measure of the sample permeability. Future work will include calibration of the ultrasound probing system to find the constant C and obtain accurate measurements of the permeability. It must be noted that the ultrasound system provides a relative measure of the line-of-sight effective permeability, due to the fact that the structure of the sample at a given height is not uniform (i.e. there is always a thin layer of unburned foam on the edges, and voids form in the char region).

The speaker-microphone pairs are positioned at the same heights on the sample centerline as the top five thermocouples: at 20 mm, 40 mm, 60 mm, 80 mm, and 100 mm from the bottom of the sample. The ultrasound signals pass through holes of 10 mm diameter on the sides of the sample holder. The speakers and microphones (Murata models MA40S4S and MA40S4R respectively) are of 10 mm diameter. In order to protect the speakers and microphones from excessive heat, the speakers and microphones are separated from the foam sample sides by copper tube waveguides, which serve as conduits for the ultrasonic signal to and from the sample. The waveguides are 50 mm long and have a 10 mm inner diameter.

The ultrasound probing system includes equipment used to produce, amplify, and channel the signals to the microphones and from the speakers. A function generator sends a six-pulse 40 kHz wavetrain, amplified to 20 V peak-to-peak, to the speaker. The signal received by the microphone is amplified and converted to a DC RMS signal, which is read at a rate of 1 Hz by the data acquisition program. The amplitude of the first peak in the RMS signal is used as the output signal because it is not affected by reflections within the sample holder and therefore provides a measure of the direct transmission through the sample.

Temperature and ultrasound output signal data from a characteristic test showing smoldering and transition to flaming show the temperature at the igniter/fuel interface (0 mm) rise when the igniter is activated. When the igniter is deactivated at the end of the ignition period, the temperatures at 0 mm and 20 mm decrease. The propagation of the smolder reaction is shown by the increasing temperatures as the smolder front approaches a thermocouple location, and subsequent drop after the smolder reaction has passed. The temperature rise, peak, and drop of the thermocouples at 40 mm and 60 mm are characteristic of a self-propagating smolder reaction. The change in slope of the temperature data at 80 mm indicates a more intense smolder reaction and secondary char oxidation at that location, and is quickly followed by a sharp increase in temperature at all thermocouples, indicating a transition to flaming event. The transition to flaming occurs at 1029 seconds, as determined by studying the temperature data in higher detail.

The output signal RMS voltage data provides a relative measure of the line-of-sight average permeability at the specified sample heights. The vertical propagation of the smolder front in the sample is shown by the sequential increases in permeability at 20 mm, 40 mm, 60 mm, and 80 mm. By comparison, these permeabilities first increase when the temperature at that location is in the range of 250 to 270ºC, corresponding to the arrival of the pyrolysis front which precedes the smolder front. The pyrolysis reaction vaporizes a portion of the fuel, causing an increase in local permeability. The permeability continues to rise as the smolder front approaches and passes each ultrasound probing

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| | location. The first peak in permeability at 40, 60, and 80 mm occurs approximately at the same time as the temperature is cresting at each location, corresponding to the passage of the smolder front. The seemingly random fluctuations in permeability occurring after the first peak are most likely due to the formation and subsequent burning of skins (i.e., melted layers of char that have been observed in samples after the test), which can impede the transmission of the ultrasound signal. As the char is oxidized (secondary char oxidation) and the skins are burnt away, voids form and grow in the char region behind the smolder front, as shown by the continued net increase in permeability after the smolder front has passed. The transition to flaming event, which occurs at 1029 seconds, is preceded by a sharp increase in the permeability at 80 mm, and then at 60 mm, indicating the presence of a strong secondary char oxidation reaction which creates large voids in the char region as it propagates downwards. The gaseous mixtures of fuel and oxidizer in the voids are heated by the char reactions. As the temperature rises, gas-phase reactions occur in the voids, leading to thermal runaway and gas-phase ignition. The transition to flaming is by very large increases in the permeability at all ultrasound probing locations, as the char and remaining foam are rapidly burned away. |
| **Task Progress:** | Smolder Velocity Experiments<br><br>Smolder velocity data is collected as well. A fuel sample is placed inside the duct and smoldered at system settings that will not allow transition to flaming to occur. A set of six thermocouples record the temperature history of the sample at evenly spaced locations along its central lengthwise axis, thus the thermocouples act to divide the fuel sample into five segments. To calculate the smolder velocity a tangent line is traced along the upward slope of each resulting temperature curve. The time at the intersection of this tangent line and a predetermined temperature (in this case 360ºC) is recorded for each thermocouple. The separation of each thermocouple (20 mm) is divided by the difference in time corresponding to two adjacent thermocouples to yield the smolder velocity for each segment. The velocities of the three central fuel segments are then averaged. Each test is repeated three times and the resulting smolder velocities are also averaged.<br><br>The dependence of smolder velocity on system parameters is measured by varying one parameter while holding all others constant. The parameters investigated include oxygen concentration, IR heat flux, and external flow velocity. Oxygen concentration is controlled by mixing either oxygen or nitrogen with both internal and external flows. Two resulting trends occurred. As expected an increase in oxygen concentration increases smolder velocity. Also a peak in smolder velocity is reached at 0.5 m/s external flow with the smolder being extinguished at 2 m/s.<br><br>Kinetics of Polyurethane for Numerical Models of Smoldering<br><br>Numerical models of smolder combustion need information on the heterogeneous reactions taking place in the thermally degrading solid. However, it is difficult to establish the smolder kinetic parameters of solids with certainty, especially for materials with complex kinetics like polymers like Polyurethane (PU). This work attempts to provide that information, by proposing a kinetic mechanism of PU smolder, estimating the needed kinetic parameters, and lastly applying them to model smoldering combustion.<br><br>The thermal degradation of PU can be approximated by a reduced mechanism consisting of only a few reactions. New thermogravimetric (TG) experiments [1] suggest that a five-step mechanism is appropriate; two foam-pyrolysis, two foam oxidations, and one char oxidation.<br><br>A mathematical model of the mass-loss is developed to simulate TG condition, assuming the reaction rates are in Arrhenius form. The 16 kinetic parameters needed are: the four activation energies , four pre-exponential factors , four reaction-order coefficients , and four stoichiometric yields . The mathematical model is used to identify these parameters that best reproduces the experimental curves of Chao et al. [1]. The solution to this inverse problem is of considerable difficulty and an efficient multidimensional optimization technique such as a Genetic Algorithm (GA) is therefore necessary. GA is a nontraditional and heuristic search method that imitates the principles of biological adaptation.<br><br>The ultimate goal of this work is to apply these parameters to the numerical modelling of smoldering. Thus a propagating smolder-front was modeled with the purpose of studying the structure and the role of each reaction. The model is one-dimensional and in steady state, solving for the species equations. The model solves the corresponding boundary value problem consisting of a system of ODEs.<br><br>In forward flow, the oxidation and the pyrolysis fronts overlap minimally to form two distinct fronts propagating downstream. In contrast, in opposed flow the oxidation and the pyrolysis fronts overlap to form one compacted front propagating upstream. These results are consistent with experimental observation, where the propagation front in forward is seen experimentally as two fronts and in opposed as one single front.<br><br>Up to now, models of smoldering of polyurethane and other materials were forced to use different kinetic schemes depending on the propagation mode. In forward, at least two reactions were needed (one pyrolysis and one oxidation) to match the dual front. In opposed, only one global reaction was used, thus changing the mechanism and the parameters. Here, it is proven that the five-reaction mechanism and one single set of parameters are able to capture the front structure in phenomenological accordance to experiments in both propagation modes. This final application further validates the kinetic parameters derived for PU and shall be of great value to numerical modeling of the process.<br><br>Reference:<br><br>1. C.Y.H. Chao, J.H. Wang, 2001, Comparison of the Thermal Decomposition Behavior of a Non-Fire Retarded and a Fire Retarded Flexible Polyurethane Foam, J. Fire Science 19, 137-155. |
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